Abstract

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Contents

Chapter 1 Introduction — Page 3

Chapter 2 Some Basics of Finite Fields — Page 4

Chapter 3 Exponential Sums over Finite Fields — Page 5

3.1 Characters of Finite Abelian Groups 5

3.2 Characters of Finite Fields 9

3.3 Gauss Sums 9

3.4 Jacobi Sums 9

3.5 L-functions 9

Chapter 4 Equations over Finite Fields — Page 10

Chapter 5 Stepanov and Bombieri's Method — Page 11

Chapter 6 Pseudopolynomials — Page 12

Chapter 7 Decoding from Quadratic Character Evaluations — Page 13

Chapter 1

Introduction

Chapter 2

Some Basics of Finite Fields

Chapter 3

Exponential Sums over Finite Fields

Exponential sums are important tools in number theory for solving problems involving integers. These sums have many applications in Analytic Number Theory.

In this chapter we will set up exponential sums for finite fields by special group homomorphisms called *characters*. Since a finite field has two different inherent group structure; one is additive and the other is multiplicative we obtain two different characters. Exponential sums are formed by using the values of one or more characters and possibly combining them with weights or with other function values.

3.1 Characters of Finite Abelian Groups

Let G be a finite abelian group with the identity element 1_G. Let S^1 be the set of complex numbers with absolute value 1. Then consider the multiplicative group S^1 with the operation of multiplication.

Definition 3.1.1: Character χ of G

A *character* χ of G is a group homomorphism $\chi : G \to S^1$ to the multiplicative group S^1 i.e. for all $g_1, g_2 \in G$

$$\chi(g_1 \cdot g_2) = \chi(g_1) \cdot \chi(g_2)$$

So if χ is any character of G then $\chi(1_G) = 1$. Since by definition for all $g \in G$,

$$\chi(g)^{|G|} = \chi(g^{|G|}) = \chi(1_G) = 1$$

the values of χ are $|G|^{th}$ roots of unity. Also we have for any $g \in G$,

$$\chi(g)\chi(g^{-1}) = \chi(g \cdot g^{-1}) = \chi(1_G) = 1$$

so $\chi(g^{-1}) = \chi(g)^{-1} = \overline{\chi(g)}$.

There are some special characters of G which have specific name. The character χ_0 defined by $\chi_0(g) = 1$ for all $g \in G$ is called the *trivial character*. For any character χ of G, the character $\overline{\chi}$ defined by $\overline{\chi}(g) = \overline{\chi(g)}$ for all $g \in G$ is called the *conjugate character* of χ.

Given any finitely many characters of G, $\chi_1, \ldots, \chi_n$ we can define the product character $\chi = \chi_1 \cdot \chi_2 \cdots \chi_n$ such that for all $g \in G$,

$$\chi(g) = \chi_1(g) \cdot \chi_2(g) \cdots \chi_n(g)$$

Hence for all $g \in G$,

$$\chi(g) \cdot \overline{\chi}(g) = |\chi(g)|^2 = 1$$

Hence the $\overline{\chi}$ is also the inverse character of χ. Hence the set of all characters of G forms a group, $\mathrm{Hom}(G, S^1)$ under multiplication whose identity element is the trivial character χ_0.

Definition 3.1.2: Dual Group to G

For a finite abelian group G, the group of all characters of G with the identity element χ_0 is called the *dual group* to G and is denoted by $\widehat{G}$ i.e. $\widehat{G} := \mathrm{Hom}(G, S^1)$.

Observation 3.1. *The dual group $\widehat{G}$ is abelian.*

Since the values of characters of G can only be $|G|^{th}$ roots of unity, the dual group $\widehat{G}$ is a finite abelian group.

Lemma 3.1.1

If G is a finite cyclic group of order n and let g be the generator of G, then for all $j \in \{0, 1, \ldots, n-1\}$ define the function

$$\chi_j(g^k) = e(j \cdot k/n) \quad \forall\, k \in \{0, 1, \ldots, n-1\}$$

Then $\widehat{G} = \{\chi_j : j \in \{0, 1, \ldots, n-1\}$.

Proof: Certainly for all $j \in \{0, 1, \ldots, n-1\}$ defines a character of G. On the other hand suppose, $\chi \in \widehat{G}$. Then $\chi(g)$ must be an nth root of unity. Hence there exists $j \in \{0, 1, \ldots, n-1\}$ such that $\chi(g) = e(j/n)$. Therefore $\chi = \chi_j$. Hence $\widehat{G} = \{\chi_j : 0 \leq j \leq n-1\}$. ■

Observation 3.2. *The characters defined in above lemma forms a cyclic group with χ_1 being the generator of $\widehat{G}$.*

Corollary 3.1.2

If G is a finite cyclic group of order n with g being the generator then $G \cong \widehat{G}$ by the isomorphism $\varphi : g^j \mapsto \chi_j$ for all $j \in \{0, 1, \ldots, n-1\}$.

Now we will show that we can extend a character of any subgroup to a character of the group.

Theorem 3.1.3

Let H be a subgroup of the finite abelian group G and let χ_H be a character of H. Then χ_H can be extended to a character of G that is there exists a character $\chi \in \widehat{G}$ with $\chi(h) = \chi_H(h)$ for all $h \in H$.

Proof: Let H be a proper subgroup of G. Let $\alpha \in G \setminus H$. Then let H_1 be the group generated by H and α i.e. $H_1 = \langle H, a \rangle$. Let m be the smallest positive integer such that $\alpha^m \in H$. Then for all $g \in H_1$, g can be written uniquely as $g = \alpha^j \cdot h$ where $0 \leq j < m$ and $h \in H$.

Let ω be a complex number such that $\omega^m = \chi_H(\alpha^m)$. Then for all $g \in H_1$ where $g = \alpha^j \cdot h$, $h \in H$ define the function $\chi_1(g) = \omega^j \chi_H(h)$. Then for any $g_1, g_2 \in H_1$ where $g_1 = \alpha^i \cdot h_1$, $g_2 = \alpha^j \cdot h_2$, we have $g_1 \cdot g_2 = \alpha^{j+k} \cdot h_1 \cdot h_2 = \alpha^{j+k \bmod m} \cdot h_1 \cdot h_2$. Hence

$$\chi_1(g_1 \cdot \chi_2) = \omega^{j+k \bmod m} \chi_H(h_1) \cdot \chi_H(h_2) = \alpha^j \chi_H(h_1) \cdot \omega^k \chi_H(h_2) = \chi_1(g_1) \cdot \chi_1(g_2)$$

Hence χ_1 is a character of H_1. And also for all $h \in H$ we have $\chi_1(h) = \chi_H(h)$.

So if $H_1 = G$ we are done. Otherwise we can continue this process above until we get G. This process we have to do finitely many times as G is a finite group. ■

Therefore if we have two distinct elements $g_1, g_2 \in G$, $g_1 \neq g_2$ then take the subgroup generated by the element $h = g_1 \cdot g_2^{-1} \neq 1$. Since this subgroup is a cyclic subgroup generated by h, by Lemma 3.1.1 we get a character χ' of the cyclic subgroup such that $\chi'(h) \neq 1$. Then by Theorem 3.1.3 we get a character χ of G such that $\chi(h) = \chi'(h) \neq 1$ and hence $\chi(g_1) \neq \chi(g_2)$. Therefore we have the following observation

Observation 3.3. *For any two distinct elements $g_1, g_2 \in G$, $g_1 \neq g_2$ we can get a character $\chi \in \widehat{G}$ such that $\chi(g_1) \neq \chi(g_2)$.*

Now consider any character χ of G. If $g \in G$ such that $\chi(g) = 1$ then $\chi(g^{-1}) = 1$. Hence the set of all elements $g \in G$ such that $\chi(g)$ forms a group. This group is called the *annihilator* of χ and is denoted by $\mathrm{Ann}(\chi)$.

Similarly suppose H is a subgroup of G. Then if $\chi \in \widehat{G}$ is a character of G such that $\chi(h) = 1$ for all $h \in H$ then $\overline{\chi}(h) = 1$ for all $h \in H$. Hence the set of all characters $\chi \in \widehat{G}$ such that $\chi(h) = 1$ for all $h \in H$ forms a group. This group is called the *annihilator* of H and is denoted by $\mathrm{Ann}(H)$. Since in both cases the notation Ann is used usually the context will make it clear whether we are talking about the annihilator of a character or the annihilator of a subgroup.

Definition 3.1.3: Annihilator

Let G be a finite abelian group. For any character $\chi \in \widehat{G}$, the *annihilator* of χ, $\mathrm{Ann}(\chi) = \{g \in G \colon \chi(g) = 1\}$ which is a subgroup of G.

For any subgroup H of G, the *annihilator* of H, $\mathrm{Ann}(H) = \{\chi \in \widehat{G} \colon \chi(h) = 1 \text{ for all } h \in H\}$ which is a subgroup of $\widehat{G}$.

Now every element of $\widehat{G}$ is a function χ which takes an element of G and maps to a complex number in S^1. Similarly we can think of every element of as a function from $\widehat{G}$ to S^1. In fact for every element $g \in G$ define the homomorphism $\mathrm{ev}_g : \widehat{G} \to S^1$ where $\mathrm{ev}_g(\chi) = \chi(g)$. Hence ev_g is a character of $\widehat{G}$ for all $g \in G$.

Theorem 3.1.4

Let G be a finite abelian group.

(i) If $\chi \in \widehat{G}$ be any nontrivial character then

$$\sum_{g \in G} \chi(g) = 0 \tag{3.1}$$

(ii) If $g \in G$ with $g \neq 1_G$ then

$$\sum_{\chi \in \widehat{G}} \chi(g) = 0 \tag{3.2}$$

Proof: Since χ is nontrivial there exists $h \in G$ such that $\chi(h) \neq 1$. Then

$$\chi(h) \sum_{g \in G} \chi(g) = \sum_{g \in G} \chi(h \cdot g) = \sum_{g \in G} \chi(g)$$

Hence we have

$$(\chi(h) - 1) \sum_{g \in G} \chi(g) = 0$$

Since we know $\chi(h) \neq 1$ we have $\sum_{g \in G} \chi(g) = 0$.

Now from the discussion above for any $g \in G$ and $\chi \in hG$ we have $\chi(g) = \mathrm{ev}_g(\chi)$. Hence

$$\sum_{\chi \in \widehat{G}} \chi(g) = \sum_{\chi \in \widehat{G}} \mathrm{ev}_g(\chi)$$

Since $g \neq 1_G$, there is a $\chi \in \widehat{G}$ such that $\chi(g) \neq 1$. Hence ev_g is a nontrivial character of $\widehat{G}$. Hence by the first part of the theorem we have $\sum_{\chi \in \widehat{G}} \mathrm{ev}_g(\chi) = 0$. ∎

The above theorem leaves the case when χ is the trivial character. Then for all $g \in G$, $\chi(g) = 1$. Hence we have $\sum_{g\in G} \chi(g) = |G|$. Hence we have the following corollary

Observation 3.4. *For any finite abelian group G. Then*

(i) Let χ is a character of G.

$$\sum_{g\in G} \chi(g) = \begin{cases} |G| & \text{if } \chi = \chi_0 \\ 0 & \text{otherwise} \end{cases} \tag{3.3}$$

(ii) $|G| = \sum_{\chi\in\widehat{G}} \sum_{g\in G} \chi(g)$

So now by counting in two ways for the sum $\sum_{\chi\in\widehat{G}} \sum_{g\in G} \chi(g)$ we get the following theorem:

Theorem 3.1.5

For any finite abelian group G, we have $|G| = |\widehat{G}|$.

Proof: From Observation 3.4(ii) we have

$$|G| = \sum_{\chi\in\widehat{G}} \sum_{g\in G} \chi(g) = \sum_{g\in G} \sum_{\chi\in\widehat{G}} \chi(g) = |\widehat{G}|$$

Hence we have the result. ■

From above theorem and Observation 3.4(i) we get the *orthogonality relations for characters.*

Observation 3.5. *Let G be a finite abelian group.*

(i) Let χ, τ are characters of G. Then

$$\frac{1}{|G|} \sum_{g\in G} \chi(g) \cdot \overline{\tau(g)} = \begin{cases} 1 & \text{if } \chi = \tau \\ 0 & \text{otherwise} \end{cases}$$

(ii) If $g, h \in G$ are any two elements then

$$\frac{1}{|G|} \sum_{\chi\in\widehat{G}} \chi(g) \cdot \overline{\chi(h)} = \begin{cases} 1 & \text{if } g = h \\ 0 & \text{otherwise} \end{cases}$$

Since we already showed that for every element $g \in G$, ev_g is a character of $\widehat{G}$. Theorem 3.1.5 gives us the possibility that G and $\widehat{G}$ are isomorphic. In Lemma 3.1.1 we already showed that for cyclic groups G and $\widehat{G}$ are isomorphic. In the next few results we will show that this is the case for all finite abelian groups.

By Fundamental Theory of Finite Abelian Groups every finite abelian group G can be written as a direct product of cyclic groups. Hence it is enough to show that if G_1 and G_2 are two finite abelian groups and $G = G_1 \times G_2$ then $\widehat{G}$ is isomorphic to the direct product of $\widehat{G}_1$ and $\widehat{G}_2$.

Theorem 3.1.6

Let G_1 and G_2 be two finite abelian groups and let $G = G_1 \times G_2$. Then $\widehat{G} \cong \widehat{G}_1 \times \widehat{G}_2$.

Proof: Since G is the direct product of G_1 and G_2, G consists of pairs (g_1, g_2) where $g_1 \in G_1$ and $g_2 \in G_2$. So consider the map $\varphi : \widehat{G}_1 \times \widehat{G}_2 \to \widehat{G}$ where for any $\chi_1 \in \widehat{G}_1$ and $\chi_2 \in \widehat{G}_2$, $\chi := \varphi(\chi_1, \chi_2)$ is the character of G defined by the following:

$$\text{for all } (g_1, g_2) \in G \qquad \chi(g_1, g_2) = \chi(g_1, 1) \cdot \chi(1, g_2) = \chi_1(g_1) \cdot \chi_2(g_2)$$

Hence φ is a homomorphism. Now we will show that φ is an isomorphism.

Let $\chi \in \widehat{G}$. Then define for all $\chi_1(g_1) = \chi(g_1, 1)$ for all $g_1 \in G_1$ and similarly define $\chi_(g_2) = \chi(1, g_2)$ for all $g_2 \in G_2$. Then χ_1 is a character of G_1 and χ_2 is a character of G_2. And also for any $(g_1, g_2) \in G$ we have

$$\chi_1(g_1) \cdot \chi_2(g_2) = \chi(g_1, 1) \cdot \chi(1, g_2) = \chi(g_1, g_2)$$

Hence $\varphi(\chi_1, \chi_2) = \chi$. Hence φ is an isomorphism. Therefore $\widehat{G} \cong \widehat{G}_1 \times \widehat{G}_2$. ■

Theorem 3.1.7

For any finite abelian group G, G is isomorphic to $\widehat{G}$.

Proof: By Fundamental Theorem of Finite Abelian Groups, G can be written as a direct product of cyclic groups i.e. there exists cyclic groups $Z_{n_1}, Z_{n_2}, \ldots, Z_{n_k}$ such that

$$G \cong Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_k}$$

Hence by repeated use of Theorem 3.1.6 we have $\widehat{G} \cong \widehat{Z}_{n_1} \times \widehat{Z}_{n_2} \times \cdots \times \widehat{Z}_{n_k}$. Since by Lemma 3.1.1 for all $i \in \{1, 2, \ldots, k\}$, $\widehat{Z}_{n_i} \cong Z_{n_i}$ we have $\widehat{G} \cong Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_k} \cong G$ Hence we have the result. ■

3.2 Characters of Finite Fields

3.3 Gauss Sums

3.4 Jacobi Sums

3.5 L-functions

Chapter 4

Equations over Finite Fields

Chapter 5

Stepanov and Bombieri's Method

Chapter 6

Pseudopolynomials

Chapter 7

Decoding from Quadratic Character Evaluations